Harnack and Poisson metrics coincide with at most one nonscalar row

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Status. Likely valid partial result, subject to human review. It gives an affirmative answer to a dimension class strictly larger than the noncommutative-ball and regular-polydisk cases stated in the source.

1 Source question

Popescu's paper *Hyperbolic geometry on noncommutative polyballs*, arXiv:1701.00766, asks on page 4 whether its Harnack metric δ_H and Poisson metric $\delta_{\mathcal{P}}$ coincide on every regular polyball [1]. The relevant source page is reproduced below; the open problem is the final paragraph before Section 1.

for any $\mathbf{A}, \mathbf{B} \in \mathbf{D}^k(\mathcal{H})^-$. Consequently, the metrics δ_H and $\delta_{\mathcal{P}}$ coincide on the Harnack (resp. Poisson) parts of $\mathbf{D}^k(\mathcal{H})^-$. We show that the hyperbolic metric δ_H is complete on the regular polydisk and the δ_H-topology coincides with the operator norm topology on $\mathbf{D}^k(\mathcal{H})$. As a consequence, $(\mathbf{D}^k(\mathcal{H}), \delta_H)$ is a complete hyperbolic space. Moreover, δ_H is invariant under the automorphism group $Aut(\mathbf{D}^k)$ and

$$\delta_H(f(\mathbf{X}), f(\mathbf{Y})) \leq \delta_H(\mathbf{X}, \mathbf{Y}), \qquad \mathbf{X}, \mathbf{Y} \in \mathbf{D}^k(\mathcal{H})$$

for any free holomorphic function $f : \mathbf{D}^k(\mathcal{H}) \to [B(\mathcal{H})^m]_1$. Therefore, the hyperbolic metric δ_H on $\mathbf{D}^k(\mathcal{H})$ has similar properties to those of the Poincaré distance on the open unit disc $\mathbb{D}$. In addition, we prove that if $\mathbf{A}$ and $\mathbf{B}$ are in $\mathbf{D}^k(\mathcal{H})$, then

$$\delta_H(\mathbf{A}, \mathbf{B}) = \ln \max \left\{ \left\| C_{\mathbf{A}}(\mathbf{R}) C_{\mathbf{B}}(\mathbf{R})^{-1} \right\|, \left\| C_{\mathbf{B}}(\mathbf{R}) C_{\mathbf{A}}(\mathbf{R})^{-1} \right\| \right\},$$

where

$$C_{\mathbf{X}}(\mathbf{R}) := (I \otimes \mathbf{\Delta_X}(I)^{1/2}) \prod_{i=1}^{k} (I - R_i \otimes X_i^*), \qquad \mathbf{X} = (X_1, \ldots, X_k) \in \mathbf{D}^k(\mathcal{H}),$$

and $R_1, \ldots R_k$ are the shift operators on the Hardy space $H^2(\mathbb{D}^k)$. In particular, we show that $\delta_H|_{\mathbb{D}^k \times \mathbb{D}^k}$ is equivalent to the Kobayashi distance on the polydisk $\mathbb{D}^k$ (see [9]) and

$$\delta_H(\mathbf{z}, \mathbf{w}) = \frac{1}{2} \ln \frac{\prod_{i=1}^{k} (1 + |\psi_{z_i}(w_i)|)}{\prod_{i=1}^{k} (1 - |\psi_{z_i}(w_i)|)}$$

for any $\mathbf{z} = (z_1, \ldots, z_k)$ and $\mathbf{w} = (w_1, \ldots, w_k)$ in $\mathbb{D}^k$, where $\psi_{\mathbf{z}} := (\psi_{z_1}, \ldots, \psi_{z_n})$ is the involutive automorphisms of $\mathbb{D}^k$ such that $\psi_{z_i}(0) = z_i$ and $\psi_{z_i}(z_i) = 0$.

We remark that, according to the results of the present paper and those from [21], the metrics δ_H and $\delta_{\mathcal{P}}$ coincide for the regular polydisk $\mathbf{D}^k(\mathcal{H})$ and for the noncommutative ball $[B(\mathcal{H})^{n_1}]_1$. It remains an open problem whether the same result is true for any regular polyball. On the other hand, it will be interesting to see to what extent these results extend to more general polydomains in $B(\mathcal{H})^m$ such as those studied in [24] and [26].

1. Harnack and Poisson equivalences on the closed polyball

In this section, we recall some basic facts concerning the noncommutative Berezin transforms on polyballs and introduce the Harnack and the Poisson equivalence relations on the closed polyball $\mathbf{B_n}(\mathcal{H})^-$. We provide a Harnack type inequality for positive free k-pluriharmonic functions and use it to show that the Harnack (resp. Poisson) equivalence class containing the zero element coincides with the open polyball $\mathbf{B_n}(\mathcal{H})$.

Let H_{n_i} be an n_i-dimensional complex Hilbert space with orthonormal basis $e_1^i, \ldots, e_{n_i}^i$. We consider the *full Fock space* of H_{n_i} defined by $F^2(H_{n_i}) := \mathbb{C}1 \oplus \bigoplus_{p \geq 1} H_{n_i}^{\otimes p}$, where $H_{n_i}^{\otimes p}$ is the (Hilbert) tensor product of p copies of H_{n_i}. Let $\mathbb{F}_{n_i}^+$ be the unital free semigroup on n_i generators $g_1^i, \ldots, g_{n_i}^i$ and the identity g_0^i. Set $e_\alpha^i := e_{j_1}^i \otimes \cdots \otimes e_{j_p}^i$ if $\alpha = g_{j_1}^i \cdots g_{j_p}^i \in \mathbb{F}_{n_i}^+$ and $e_{g_0^i}^i := 1 \in \mathbb{C}$. The length of $\alpha \in \mathbb{F}_{n_i}^+$ is defined by $|\alpha| := 0$ if $\alpha = g_0^i$ and $|\alpha| := p$ if $\alpha = g_{j_1}^i \cdots g_{j_p}^i$, where $j_1, \ldots, j_p \in \{1, \ldots, n_i\}$. We define the *left creation operator* $S_{i,j}$ acting on the Fock space $F^2(H_{n_i})$ by setting $S_{i,j} e_\alpha^i := e_j^i \otimes e_\alpha^i$, $\alpha \in \mathbb{F}_{n_i}^+$, and the operator $\mathbf{S}_{i,j}$ acting on the Hilbert tensor product $F^2(H_{n_1}) \otimes \cdots \otimes F^2(H_{n_k})$ by setting

$$\mathbf{S}_{i,j} := \underbrace{I \otimes \cdots \otimes I}_{i-1 \text{ times}} \otimes S_{i,j} \otimes \underbrace{I \otimes \cdots \otimes I}_{k-i \text{ times}},$$

where $i \in \{1, \ldots, k\}$ and $j \in \{1, \ldots, n_i\}$. We denote $\mathbf{S} := (\mathbf{S}_1, \ldots, \mathbf{S}_k)$, where $\mathbf{S}_i := (\mathbf{S}_{i,1}, \ldots, \mathbf{S}_{i,n_i})$, or $\mathbf{S} := \{\mathbf{S}_{i,j}\}$. The noncommutative Hardy algebra $\mathbf{F_n^\infty}$ (resp. the polyball algebra $\boldsymbol{\mathcal{A}}_\mathbf{n}$) is the weakly closed (resp. norm closed) non-selfadjoint algebra generated by $\{\mathbf{S}_{i,j}\}$ and the identity. Similarly, we define the *right creation operator* $R_{i,j} : F^2(H_{n_i}) \to F^2(H_{n_i})$ by setting $R_{i,j} e_\alpha^i := e_\alpha^i \otimes e_j^i$ for $\alpha \in \mathbb{F}_{n_i}^+$, and the operator $\mathbf{R}_{i,j}$ acting on the Hilbert tensor product $F^2(H_{n_1}) \otimes \cdots \otimes F^2(H_{n_k})$ by setting

$$\mathbf{R}_{i,j} := \underbrace{I \otimes \cdots \otimes I}_{i-1 \text{ times}} \otimes R_{i,j} \otimes \underbrace{I \otimes \cdots \otimes I}_{k-i \text{ times}}.$$

Transcription. "It remains an open problem whether the same result is true for any regular polyball." Here "the same result" is $\delta_H = \delta_{\mathcal{P}}$, known there for a noncommutative ball and for a regular polydisk.

2 Definitions and statement

Let $\mathbf{B_n}$ be the regular polyball with $\mathbf{n} = (n_1, \ldots, n_k)$, and let $\boldsymbol{\mathcal{P}}(\mathbf{R}, \mathbf{X})$ be its free pluriharmonic Poisson kernel. For $\mathbf{A}, \mathbf{B} \in \mathbf{B_n}(\mathcal{H})^-$ and $c > 1$, write $\mathbf{A} \sim_{\mathcal{P},c} \mathbf{B}$ when

$$c^{-2}\boldsymbol{\mathcal{P}}(\mathbf{R}, r\mathbf{B}) \le \boldsymbol{\mathcal{P}}(\mathbf{R}, r\mathbf{A}) \le c^2\boldsymbol{\mathcal{P}}(\mathbf{R}, r\mathbf{B}), \qquad 0 \le r < 1. \tag{1}$$

Write $\mathbf{A} \sim_{H,c} \mathbf{B}$ when the same inequalities hold after replacing the Poisson kernel by every positive operator-valued free k-pluriharmonic function F. The metrics are the logarithms of the infima of the admissible constants on their respective equivalence parts.

Theorem 1. *Suppose that at most one coordinate of* $\mathbf{n}$ *is larger than one. For every* $\mathbf{A}, \mathbf{B} \in \mathbf{B_n}(\mathcal{H})^-$ *and every* $c > 1$,

$$\mathbf{A} \sim_{H,c} \mathbf{B} \quad \Longleftrightarrow \quad \mathbf{A} \sim_{\mathcal{P},c} \mathbf{B}.$$

Consequently the Harnack parts and Poisson parts coincide, and

$$\delta_H(\mathbf{A}, \mathbf{B}) = \delta_{\mathcal{P}}(\mathbf{A}, \mathbf{B})$$

on every such part.

3 Proof intuition

One implication is built into the definitions: the Poisson kernel is itself a positive free pluriharmonic function. For the reverse implication, the key input is that with only one genuinely free row, every positive free pluriharmonic function is a completely positive image of the universal Poisson kernel. A completely positive map preserves both inequalities in (1), and it does not enlarge the constant. Thus the two sets of admissible constants are identical, not merely comparable.

The one-row restriction enters only in the representation lemma below. Scalar rows are commuting isometries and can be made unitary simultaneously by an $\mathbb{N}^s$ direct limit. Fuglede's theorem then supplies the missing adjoint commutation with the one nonscalar row.

4 The representation input

Lemma 2. *If at most one* n_i *is larger than one, every positive free* k*-pluriharmonic function* F *on* $\mathbf{B_n}$*, with coefficients in* $B(\mathcal{E})$*, has the form*

$$F(\mathbf{X}) = (\mu \otimes \mathrm{id})[\boldsymbol{\mathcal{P}}(\mathbf{R}, \mathbf{X})] \tag{2}$$

for a completely positive map $\mu : C^*(\mathbf{R}) \to B(\mathcal{E})$.

Proof. First normalize $F(0) = I_{\mathcal{E}}$. The positive-structure theorem of [2] gives $\mathcal{K} \supset \mathcal{E}$ and mutually commuting row isometries $\mathbf{V} = (V_1, \ldots, V_k)$ whose compressed mixed moments are the coefficients of F.

After relabeling, V_1 is the possible nonscalar row and $V_2, \ldots, V_k$ are commuting scalar isometries $U_1, \ldots, U_s$. For $p \le q$ in $\mathbb{N}^s$, connect a copy $\mathcal{K}_p$ of $\mathcal{K}$ to $\mathcal{K}_q$ by the isometry U^{q-p}. On the Hilbert direct limit, define

$$\widetilde{U}_i J_p h = J_p U_i h, \qquad \widetilde{V}_j J_p h = J_p V_{1,j} h.$$

These definitions are compatible with the connecting maps. Each $\widetilde{U}_i$ is onto because $\widetilde{U}_i J_{p+e_i} h = J_p h$, hence is unitary. The $\widetilde{V}_j$ remain isometries with orthogonal ranges. All extensions commute; Fuglede's theorem [3] makes every $\widetilde{V}_j$ commute with every $\widetilde{U}_i^*$. Thus the extended rows doubly commute.

The universal property of the tensor product of the Cuntz–Toeplitz algebras now gives a $*$-representation $\pi : C^*(\mathbf{R}) \to B(\widetilde{\mathcal{K}})$ with $\pi(\mathbf{R})$ equal to the extended tuple. Compression to $\mathcal{E}$ defines the completely positive map $\mu(T) = J_{\mathcal{E}}^* \pi(T) J_{\mathcal{E}}$. Its moments are the coefficients of F, so coefficient uniqueness yields (2).

For general $A = F(0) \ge 0$, apply the normalized argument to

$$(A + \varepsilon I)^{-1/2}(F + \varepsilon I)(A + \varepsilon I)^{-1/2}.$$

After conjugating back, the representing maps have norms $\|A + \varepsilon I\|$. A point-ultraweak cluster limit as $\varepsilon \downarrow 0$ is completely positive and has the coefficients of F, proving (2). $\square$

5 Proof of Theorem 1

If $\mathbf{A} \sim_{H,c} \mathbf{B}$, apply the defining Harnack inequalities to the positive free pluriharmonic function $\mathbf{X} \mapsto \boldsymbol{\mathcal{P}}(\mathbf{R}, \mathbf{X})$. This gives (1), hence $\mathbf{A} \sim_{\mathcal{P},c} \mathbf{B}$.

Conversely assume (1), and let F be any positive free k-pluriharmonic function with coefficients in $B(\mathcal{E})$. By Lemma 2, choose a completely positive μ satisfying (2). Apply the completely positive map $\mu \otimes \mathrm{id}_{B(\mathcal{H})}$ to all three terms of (1). For every $0 \leq r < 1$ this gives

$$c^{-2} F(r\mathbf{B}) \leq F(r\mathbf{A}) \leq c^2 F(r\mathbf{B}).$$

Thus $\mathbf{A} \sim_{H,c} \mathbf{B}$. The admissible constants are identical in the two definitions, so their infima, their parts, and the resulting logarithmic metrics are identical. $\square$

6 Verification, novelty, and limitations

The proof is order-theoretic after Lemma 2: no limiting argument changes c. The representation lemma was independently packaged and proof-checked earlier in this run; it is repeated here so that the present packet is self-contained. The human reviewer should focus on the direct-limit orientation, the use of Fuglede to obtain double commutation, and the tensor order in (2).

A bounded search through 26 August 2026 used the exact source sentence, the paper title and arXiv id, and combinations of “Harnack”, “Poisson”, “regular polyball”, and “one nonscalar row”. It found the source, later applications, and a recent survey, but no resolution of the exact question or this dimension class. The novelty verdict is therefore “apparently new within the bounded search”.

The theorem does not settle polyballs with two or more nonscalar rows. The missing dilation would have to extend two commuting free row isometries while preserving both orthogonal-range relations and cross-adjoint commutation; ordinary $\mathbb{N}^s$ direct limits do not provide that extension.

References

[1] G. Popescu, *Hyperbolic geometry on noncommutative polyballs*, J. Math. Anal. Appl. **456** (2017), 576–607; arXiv:1701.00766.

[2] G. Popescu, *Free pluriharmonic functions on noncommutative polyballs*, Anal. PDE **9** (2016), 1185–1234; arXiv:1511.08852.

[3] J. B. Conway, *A Course in Functional Analysis*, second edition, Graduate Texts in Mathematics 96, Springer, 1990. See the Fuglede–Putnam theorem.